

HOANG TUNG (PATRICK) PHAM

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EDUCATION

Ph.D. in Statistical Science Duke University	August 2025 – Present Durham, NC, USA
BSc in Mathematics with Statistics Imperial College London, <i>First Class Honours</i>	October 2020 – June 2023 London, UK

SKILLS

Technical	Bayesian statistics, statistical theory, optimization, statistical modeling machine learning, deep learning
Programming	Python, R, Julia, SQL, MATLAB
Libraries	Pandas, NumPy, SciPy, Matplotlib, Seaborn, scikit-learn, PyTorch, tidyverse
Bayesian Software	Stan, JAGS
Languages	Vietnamese (native), English (fluent)

WORK EXPERIENCE

An Binh Joint Stock Commercial Bank <i>Junior Data Scientist</i>	October 2024 - May 2025 <i>Hanoi, Vietnam</i>
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- Developed machine learning models for customer segmentation and churn prediction using gradient boosting and XGBoost on large-scale transactional data.
- Designed hybrid rule-based and predictive systems to support personalized banking strategies and targeted marketing campaigns.
- Performed feature engineering and model optimization, achieving 85% prediction accuracy and contributing to a 12% increase in campaign conversion and 8% reduction in monthly churn.

FPT Software <i>Junior Data Engineer</i>	June 2024 - October 2024 <i>Hanoi, Vietnam</i>
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- Designed and implemented a graph-based vaccine recommendation system using Neo4j and Cypher, incorporating collaborative filtering for personalized vaccine scheduling.
- Developed patient data models and automated ETL pipelines to integrate large-scale pharmaceutical datasets into a graph database.
- Improved recommendation accuracy by leveraging demographic and medical history features for personalized vaccination planning.

Vingroup Big Data Institute <i>AI Engineer Trainee</i>	July 2023 - May 2024 <i>Hanoi, Vietnam</i>
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- Completed advanced training in mathematics and machine learning, followed by an ML research internship with the Antimicrobial Resistance (AMR) team at GeneStory.
- Applied statistical and ML methods to GWAS data to compute Polygenic Risk Scores (PRS) for thyroid cancer patients in Vietnam.
- Extended Non-negative Matrix Factorization (NMF) to single-cell multimodal omics data, developing an optimization-based unified embedding model for cell clustering.

STATISTICS AND DATA SCIENCE PROJECTS

Driver Drowsiness Detection <i>Vingroup's AI Engineer Training Program</i>
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- Used MTCNN, FaceMesh from Mediapipe, Eye Aspect Ratio (EAR) to extract and vectorize important facial features from drivers' videos
- Implemented various deep learning models such as Inception Resnet combined with LSTM, or EAR with CNN to predict drivers' drowsiness with good accuracy and recall scores (better than human judgement of 50%)

Estimating parameters with and without stratification:

A theoretical comparison

May 2023 - July 2023

Supervisor: Dr. Heather Battey, Imperial College London

London, UK

- Analyzed and compared OLS and stratified regression estimators for a target parameter in a medical statistics setting.
- Derived likelihood functions, computed estimators, and studied variance and asymptotic properties via simulation and theoretical analysis.
- Established that, under normality, the stratified estimator is asymptotically more efficient.

Modelling Santander Cycle Hires using Self-Exciting and Mutually-Exciting Point Processes

May 2022 - June 2022

Supervisor: Dr. Francesco Sanna Passino, Imperial College London

London, UK

- Developed and compared Poisson and Hawkes point process models for bike-sharing event data across 840 London stations.
- Assessed model fit via likelihood-based inference and Kolmogorov–Smirnov diagnostics.
- Demonstrated that a mutually- and self-exciting Hawkes specification provided the best representation of event dynamics.

TEACHING/TUTORING EXPERIENCE

Duke University

January 2026 - Present

Teaching Assistant, Department of Statistical Science

Durham, NC, USA

- Teaching Assistant, STA 199 (Introduction to Data Science), Duke University.
- Led weekly lab sessions and held office hours to support students with statistical concepts, R programming, and data analysis.
- Graded assignments and projects, providing detailed feedback to reinforce statistical reasoning and coding practices.

American Study

October 2024 - May 2025

Academic Tutor

Hanoi, Vietnam

- Tutored Mathematics and English for standardized exams (IELTS, SAT, AP Statistics, AP Calculus AB/BC, IGCSE, GMAT, GRE).
- Mentored students in academic planning and university applications, providing strategic guidance for competitive admissions.

DECLARATION

I hereby declare that all the details furnished above are true to the best of my knowledge and belief.